

Hyper-V Migration

Migrate from Microsoft Hyper-V to KVM/KubeVirt with hyper2kvm + hypersdk

Move Windows and Linux VMs from Hyper-V to open-source KVM infrastructure. Automatic Hyper-V Integration Components removal, VirtIO driver injection, VHD/VHDX to QCOW2 conversion, and boot verification. Eliminate Windows Server Datacenter licensing overhead and gain platform independence.

VHD/VHDX Native | Hyper-V IC Removal | VirtIO Auto-Inject | \$0 License | Apache 2.0

Why Leave Hyper-V?

Hyper-V isn't just a hypervisor cost — it's a Windows Server Datacenter licensing trap.

\$6,155

Windows Server 2022
Datacenter per 2-core pack

\$48K+

Per host (16 cores)
Datacenter licensing

\$0

KVM hypervisor cost
(built into Linux kernel)

70%

Typical cost reduction
Hyper-V → KVM migration

The Licensing Problem

- **Windows Server Datacenter:** Required for unlimited VMs per host — \$6,155 per 2-core pack, minimum 8 packs per server = **\$49,240 per host**
- **Windows Server Standard:** Only 2 VMs per license — need multiple licenses for more VMs
- **System Center:** Additional \$3,607 per 2-core pack for management
- **SQL Server on Hyper-V:** Must license all physical cores on host
- **CALs:** Per-user or per-device Client Access Licenses on top
- **Total:** A 4-host Hyper-V cluster can cost **\$200K+ in Microsoft licensing alone**

The Technical Limitations

- **Windows-only hosts:** Hyper-V requires Windows Server OS on the host
- **SCVMM complexity:** System Center Virtual Machine Manager is expensive and complex
- **Limited Linux support:** Linux VMs work but are second-class citizens
- **No KubeVirt path:** No native container/VM convergence
- **Vendor lock-in:** VHD/VHDX format, Hyper-V IC dependencies, Microsoft ecosystem
- **No edge story:** Windows Server is too heavy for edge deployments

Hyper-V (Before)

- \$49K+ per host (Datacenter licensing)
- Windows Server OS required on every host
- System Center \$28K+ per host for management
- VHD/VHDX proprietary format

KVM (After)

- \$0 per host (KVM in Linux kernel)
- Any Linux distro (RHEL, Fedora, Ubuntu)
- \$0 management (libvirt, Cockpit, virsh)
- QCOW2 open format

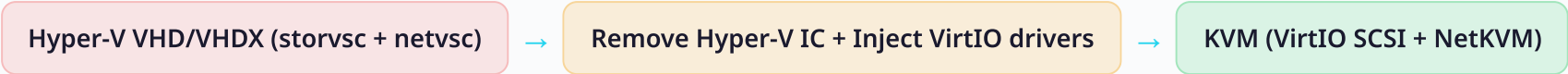
- Hyper-V IC required in every guest
- SCVMM or PowerShell management only
- No container convergence
- Heavy footprint (Windows Server host OS)

- VirtIO — open, fast, standard
- CLI + TUI + API + web management
- KubeVirt for VM + container convergence
- Lightweight — Linux host uses minimal resources

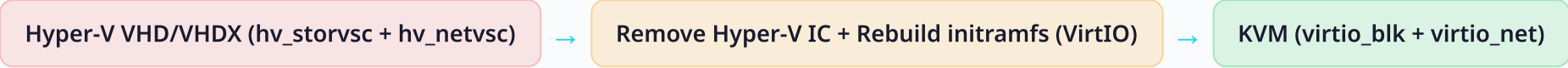
Migration Paths

hyper2kvm handles both Windows and Linux VMs from Hyper-V with source-specific fixes.

Windows VMs on Hyper-V → KVM



Linux VMs on Hyper-V → KVM



Azure VMs → KVM (Hyper-V under the hood)



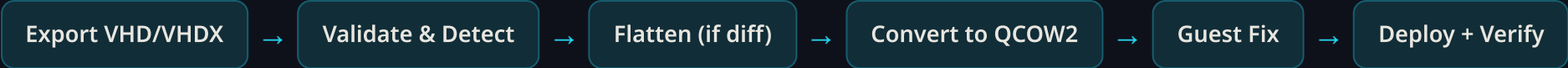
What hyper2kvm Fixes — Hyper-V Specific

Component	Hyper-V Original	KVM Replacement	hyper2kvm Action
Storage controller	storvsc (Hyper-V synthetic)	virtio-blk / virtio-scsi	Inject VirtIO storage driver + registry/initramfs fix
Network adapter	netvsc (Hyper-V synthetic)	virtio-net (NetKVM)	Inject VirtIO network driver + clean NIC configs
Memory balloon	Dynamic Memory (Hyper-V)	virtio-balloon	Install balloon driver for KVM memory management
Integration Services	hv_kvp_daemon, hv_vss_daemon, hv_fc copy_daemon	qemu-guest-agent (optional)	Remove all Hyper-V IC daemons and services
Heartbeat	hv_heartbeat	Not needed	Disable heartbeat driver and service
Time sync	hv_timesource	kvm-clock	Remove Hyper-V time source, KVM clock auto-detected

Disk format	VHD / VHDX	QCOW2	Convert VHD/VHDX → QCOW2 via qemu-img
Boot config (Windows)	BCD for storvsc	BCD for VirtIO	Update BCD entries for VirtIO disk controller
initramfs (Linux)	Contains hv_storvsc, hv_netvsc	Needs virtio_blk, virtio_net	Rebuild initramfs with VirtIO modules via dracut/update-initramfs
fstab	/dev/sda references	UUID-based mounts	Convert all entries to UUID for device-name independence

VHD/VHDX Conversion Pipeline

Native VHD and VHDX support — including dynamic, fixed, and differencing disks.



VHD/VHDX Format Support

Format	Description	Max Size	hyper2kvm Support	Notes
VHD Fixed	Pre-allocated, full size on disk	2 TB	Full support	Fastest conversion — direct mapping
VHD Dynamic	Grows as data is written	2 TB	Full support	Sparse conversion — only used blocks
VHD Differencing	Delta from parent VHD	2 TB	Full support	Auto-flatten to single image
VHDX Fixed	Pre-allocated, modern format	64 TB	Full support	Larger disk support than VHD
VHDX Dynamic	Grows as data is written	64 TB	Full support	Most common Hyper-V format
VHDX Differencing	Delta from parent (checkpoints)	64 TB	Full support	Checkpoint chains auto-flattened

Export Methods

Method 1: Direct Copy (Simplest)

- Shut down VM on Hyper-V
- Copy VHD/VHDX file from Hyper-V storage
- Default location: `C:\Users\Public\Documents\Hyper-V\Virtual Hard Disks\`
- Transfer via SMB share, SCP, or USB drive
- Works for any Hyper-V version
- **Best for:** Small number of VMs, simple setup

Method 2: Hyper-V Export

- Right-click VM in Hyper-V Manager → Export
- Exports VM config + VHD/VHDX + snapshots
- PowerShell: `Export-VM -Name "vm01" -Path "D:\exports"`
- Preserves checkpoint chain
- hyper2kvm auto-flattens checkpoints
- **Best for:** VMs with checkpoints/snapshots

Method 3: hypersdk (Automated)

- hypersdk connects to Hyper-V via WinRM or SSH
- Discovers VMs with metadata (CPU, RAM, disks)
- Exports VHD/VHDX programmatically
- Streaming progress with ETA
- Batch export multiple VMs
- **Best for:** Large estates, automated pipelines

Method 4: Azure VM Export

- Azure VMs use Hyper-V internally
- hypersdk exports via managed disk snapshot → SAS URL
- Downloads VHD format
- Same Hyper-V IC removal + VirtIO injection
- Plus: removes Azure waagent
- **Best for:** Azure-to-on-prem repatriation

YAML Configuration

```
# hyperv-linux.yaml – Linux VM from Hyper-V
cmd: local
vhd: /exports/ubuntu-web-01.vhdx
output_dir: ./converted
out_format: qcow2
flatten: true
compress: true
regen_initramfs: true      # Rebuild with VirtIO, remove hv_* modules
emit_domain_xml: true
vm_name: ubuntu-web-01
memory: 4096
vcpus: 2
machine: q35
boot_test: true
```

Windows on Hyper-V → KVM

The most common Hyper-V migration scenario — Windows Server VMs moving to KVM.

Without VirtIO injection, Windows VMs from Hyper-V will BSOD on KVM

Windows boots with `storvsc.sys` (Hyper-V synthetic storage). KVM uses VirtIO. Without pre-injecting VirtIO drivers into the offline disk, Windows can't see the boot volume and crashes with **INACCESSIBLE_BOOT_DEVICE (0x7B)**. hyper2kvm handles this automatically.

Windows Hyper-V → KVM Fix Pipeline

#	Step	What hyper2kvm Does	Why It's Needed
1	Mount disk offline	qemu-nbd attaches VHDX, mounts Windows partition	All fixes applied without booting the broken VM
2	Detect Windows version	Reads registry: CurrentVersion, CurrentBuild	Selects correct VirtIO driver set for the OS
3	Inject VirtIO storage	Copies viostor.sys/vioscsi.sys + .inf, sets Start=0 in registry	VM can see boot disk on KVM (prevents 0x7B BSOD)
4	Inject VirtIO network	Copies NetKVM.sys + .inf, adds CriticalDeviceDatabase entries	VM has network connectivity after boot
5	Inject VirtIO balloon	Copies balloon.sys + .inf for memory management	Dynamic memory works on KVM
6	Remove Hyper-V IC	Disables storvsc, netvsc, hv_kvp, hv_vss, hv_fc copy services in registry	Prevents error spam and conflicts on KVM
7	Fix BCD	Updates Boot Configuration Data for VirtIO controller	Windows boot manager finds boot volume
8	Enable serial console	Configures COM1 for headless access	Remote management without GUI
9	Generate libvirt XML	Q35 machine, VirtIO devices, UEFI if needed, TPM if Win11	Optimized VM config for KVM
10	Boot verification	Starts VM, confirms login screen (no BSOD)	Catch issues before production cutover

Supported Windows Versions from Hyper-V

OS	Editions	Boot	Status	Notes
Windows Server 2025	Standard, Datacenter	UEFI	Tested	Latest VirtIO, Q35 recommended

Windows Server 2022	Standard, Datacenter, Core	UEFI, BIOS	Tested	Most common Hyper-V target
Windows Server 2019	Standard, Datacenter	UEFI, BIOS	Tested	Widely deployed on Hyper-V
Windows Server 2016	Standard, Datacenter	UEFI, BIOS	Tested	Extended support until 2027
Windows 11	Pro, Enterprise	UEFI+TPM	Tested	TPM emulated via swtpm on KVM
Windows 10	Pro, Enterprise, LTSC	UEFI, BIOS	Tested	VDI and workstation migrations

```
# win-hyperv.yaml – Windows Server from Hyper-V
cmd: local
vhd: /exports/win-dc-01.vhdx
output_dir: ./converted
out_format: qcow2
windows: true
guest_os: windows
virtio_win_iso: /usr/share/virtio-win/virtio-win.iso
win_stage: bootstrap
emit_domain_xml: true
vm_name: win-dc-01
memory: 8192
vcpus: 4
machine: q35
uefi: true
boot_test: true
```

Linux on Hyper-V → KVM

Linux VMs on Hyper-V use hv_* kernel modules. hyper2kvm swaps them for VirtIO automatically.

Linux Hyper-V → KVM Fix Pipeline

#	Fix	What hyper2kvm Does	Technical Detail
1	initramfs rebuild	Replaces hv_storvsc/hv_netvsc with virtio_blk/virtio_scsi/virtio_net	Chroot: dracut --add-drivers "virtio_blk virtio_scsi virtio_net virtio_pci" or update-initramfs
2	fstab UUID repair	Converts /dev/sda entries to UUID-based mounts	Reads blkid, replaces device paths in /etc/fstab
3	GRUB update	Fixes root= parameter for KVM device naming	Updates /etc/default/grub, regenerates grub.cfg
4	Hyper-V IC removal	Disables hv_kvp_daemon, hv_vss_daemon, hv_fcoppy_daemon	systemctl disable hv_kvp_daemon hv_vss_daemon hv_fcoppy_daemon
5	Network cleanup	Removes Hyper-V netvsc NIC configs	Cleans ifcfg-*/netplan with Hyper-V MAC/device names, removes 70-persistent-net.rules
6	SELinux relabel	Triggers full filesystem relabel (RHEL/Fedora/Rocky)	Creates /.autorelabel for restorecon on first boot
7	Serial console	Enables ttyS0 for headless access	Adds console=ttyS0,115200n8 to GRUB, enables serial-getty@ttyS0

Supported Linux Distros from Hyper-V

RHEL Family

- RHEL 7, 8, 9, 10
- Rocky Linux 8, 9
- AlmaLinux 8, 9
- CentOS Stream 8, 9
- Oracle Linux 7, 8, 9
- Fedora 38-41
- **initramfs tool:** dracut

Debian Family

- Ubuntu 20.04, 22.04, 24.04
- Debian 11, 12
- **initramfs tool:** update-initramfs
- Most common Linux on Hyper-V
- Microsoft provides optimized kernels
- hyper2kvm replaces MS kernel config

SUSE Family & Others

- SLES 12 SP5, 15 SP5/SP6
- openSUSE Leap 15.5/15.6
- **initramfs tool:** dracut
- Btrfs subvolume-aware
- Amazon Linux 2, 2023
- All distros auto-detected

```
# linux-hyperv.yaml – Ubuntu from Hyper-V
cmd: local
vhd: /exports/ubuntu-db-01.vhdx
output_dir: ./converted
out_format: qcow2
flatten: true
compress: true
regen_initramfs: true      # Swap hv_* for VirtIO modules
emit_domain_xml: true
deploy_k8s: true          # Also generate KubeVirt manifest
vm_name: ubuntu-db-01
memory: 16384
vcpus: 8
machine: q35
boot_test: true
```

Cost Savings

Eliminating Windows Server Datacenter licensing is the biggest win.

4-Host Hyper-V Cluster → KVM — 3-Year TCO

Cost Item	Hyper-V Cluster	KVM Cluster	Savings
Windows Server Datacenter	\$196,960 (4 hosts x \$49K)	\$0 (Linux is free)	\$196,960
System Center (SCVMM)	\$57,712 (4 hosts x \$14K)	\$0 (libvirt/Cockpit)	\$57,712
Windows Server CALs	\$45,000 (500 users x \$30/ea x 3yr)	\$0	\$45,000
Software Assurance / renewal	\$90,000 (3 yr renewals)	\$0	\$90,000
Hardware (servers)	\$80,000	\$80,000 (same hardware)	\$0
OS licensing (host)	Included in Datacenter	\$0 (RHEL free) or \$6K (RHEL sub)	-\$6,000
Migration (one-time)	N/A	\$15,000 (labor)	-\$15,000
Operations staff	\$300,000 (3 yr)	\$270,000 (KVM simpler)	\$30,000
3-Year Total	\$769,672	\$371,000	\$398,672

Additional Savings Opportunities

SQL Server Licensing

SQL on Hyper-V: must license all physical cores.
SQL on KVM with CPU pinning: license only assigned cores. **20-40% SQL license reduction** — potentially \$200K+ for large SQL estates.

Windows Guest Licensing

Windows guest VMs still need licenses on KVM — but you eliminate the **host Datacenter license (\$49K/host)** which was the most expensive component. Guest Standard licenses are much cheaper.

Linux Guest Savings

Linux VMs on Hyper-V required Windows Server Datacenter on the host. On KVM, Linux runs on Linux — **zero Microsoft licensing** for pure-Linux workloads. 100% savings on host licensing.

\$399K Saved — 4-Host Cluster, 3 Years

Scale linearly: 8 hosts = **\$800K saved**. 16 hosts = **\$1.6M saved**.
The migration tool is free. The savings start the day you cancel Microsoft licenses.

Enterprise Workloads & Scenarios

Common Hyper-V workloads migrated safely with hyper2kvm.

Active Directory Domain Controllers

- NTDS.dit and SYSVOL migrate with disk
- Migrate secondary DCs first, verify replication
- VirtIO injection handles boot on KVM
- DNS zones and Kerberos keytabs intact
- **Critical:** Never migrate all DCs simultaneously

SQL Server

- Database files migrate with disk — no reinstall
- VirtIO SCSI provides excellent I/O performance
- KVM CPU pinning reduces per-core licensing
- Always-On AG: migrate secondary replicas first
- **Savings:** Eliminate Datacenter + reduce SQL cores

File Servers / DFS

- NTFS permissions (ACLs) fully preserved
- DFS namespaces and replication intact
- SMB shares auto-start after boot
- Deduplication volumes migrate correctly
- **Simplest migration** — standard pipeline

Linux Web/DB on Hyper-V

- Apache/Nginx, PostgreSQL/MySQL — standard fixes
- initramfs rebuild + fstab + GRUB fix
- Remove Hyper-V IC daemons
- **Bonus:** Move host OS from Windows to Linux too
- Zero Microsoft licensing for pure-Linux workloads

Hyper-V Migration Tool Comparison

Capability	hyper2kvm	virt-v2v	MTV	Manual	DVMS
VHD/VHDX conversion	✓ Native	✓	Partial	qemu-img	✓
Hyper-V IC removal	✓ Auto	✓	×	Manual	Partial
VirtIO injection (Windows)	✓ Auto	✓	Limited	Manual	Partial
initramfs rebuild (Linux)	✓ Auto	✓	Limited	Manual	×
Boot verification	✓ Auto	×	×	Manual	×
Deploy to KVM/libvirt	✓	✓	×	Manual	Partial

Deploy to KubeVirt	✓	×	✓	Complex	×
Interactive TUI	✓ zkvm	×	Web UI	×	×
Azure VM support	✓	×	×	Manual	×
Batch / scheduling	✓	×	✓	×	×
No libguestfs needed	✓	Required	Required	N/A	N/A
Cost	\$0	\$0	\$50K+	\$0	\$\$

Real-World Scenarios

Enterprise Hyper-V migration projects powered by hyper2kvm.

Scenario 1: Mid-Size Business — 40 VMs

- **Source:** 2-host Hyper-V cluster, Windows Server 2019 Datacenter
- **VMs:** 25 Windows (AD, SQL, IIS, file), 15 Linux (web, DB, Docker)
- **Target:** 2-host KVM cluster on RHEL 9
- **Process:** Export VHDX, hyper2kvm converts + fixes, boot verified
- **Timeline:** 3 weeks
- **Savings:** \$200K/yr (Datacenter + SCVMM + CALs eliminated)

Scenario 2: Hyper-V to KubeVirt — 30 VMs

- **Source:** Hyper-V 2019 cluster, mixed Windows/Linux
- **VMs:** VDI desktops, RDS session hosts, app servers
- **Target:** K3s + KubeVirt (modernization initiative)
- **Process:** VHDX export, Hyper-V IC removal, VirtIO injection, KubeVirt manifest generation
- **Timeline:** 2 weeks
- **Savings:** \$150K/yr + path to cloud-native

Scenario 3: Azure Repatriation — 20 VMs

- **Source:** Azure VMs (Hyper-V based internally)
- **VMs:** .NET web apps, SQL Server, file shares
- **Target:** On-prem KVM (data sovereignty)
- **Process:** hypersdk exports managed disks (VHD), hyper2kvm removes waagent + Hyper-V IC, injects VirtIO
- **Timeline:** 2 weeks
- **Savings:** \$480K/yr in Azure compute + egress

Scenario 4: Education — 100 VMs, Budget-Constrained

- **Source:** 4-host Hyper-V cluster, edu licensing
- **VMs:** Student VDI, learning management, library, email
- **Target:** KVM + KubeVirt on existing hardware
- **Process:** Batch export, automated conversion, dual target deploy
- **Timeline:** 4 weeks (summer break)
- **Savings:** \$120K/yr redirected to educational programs

Hyper-V Migration Made Simple

Native VHD/VHDX support. Automatic Hyper-V IC removal. VirtIO injection for Windows and Linux.

Eliminate Windows Server Datacenter licensing. Deploy to KVM and/or KubeVirt.

Azure VMs included (same Hyper-V stack under the hood).

hyper2kvm: the complete Hyper-V exit toolkit. Free forever.

